Eugene Mwangi Mwaniki

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SUMMARY

Electrical and Information Engineering graduate from the University of Nairobi with a strong foundation in full-stack development. Proficient in Python (Django, Flask) and JavaScript (React.js, Node.js) ecosystems with hands-on experience in building, testing, and deploying scalable applications. Eager to apply skills in microservices, cloud APIs, and AI integration to a challenging software engineering role. Ready to work in fulltime and contractor roles.

EDUCATION & CERTIFICATION

- Bachelor's Degree in Electrical and Electronics Engineering | University of Nairobi
- Meta Front End Development | Coursera
- Artificial Intelligence with Python and TensorFlow | DeepLearning.AI (Coursera)
- AWS Cloud Solutions Architect | Coursera

SKILLS

- **Languages:** Python, JavaScript, TypeScript, C, C++, Java, Kotlin, HTML5, CSS3, SQL
- **Frameworks & Libraries:**
  - **Backend:** Node.js, Express, Django, Flask, FastAPI
  - **Frontend:** React.js, Next.js, Angular, Redux
- **Databases:** MySQL, PostgreSQL, SQLite, MongoDB, NoSQL, Redis
- **Tools & Platforms:** Git, GitHub, GitLab, Docker, Amazon Web Services (AWS), Google Cloud API, OpenAI API
- **Core Concepts:** Backend Development, Frontend Development, RESTful APIs, Microservices, Agile Methodologies, Operating Systems, Android App Development, Unit Testing (Jest, Mocha, Pytest, Unittest)
- **Design:** Figma

WORK EXPERIENCE

Turing Enterprises | AI trainer (1.5 years)

- Trained and fine-tuned large language models (LLMs) on Python code to generate functional backend systems using Flask, Django, and FastAPI.
- Improved AI model proficiency in JavaScript by training it to produce high-quality, scalable code for frontend (React.js) and backend (Node.js) applications.
- Developed and refined tool-calling capabilities for AI agents, integrating real-time data APIs for weather, mapping, and financial stock information.

Octavia Carbon | Intern (6 months)

- Developed C++ firmware for direct air carbon capture machines designed to capture 1000 tons of CO_2 annually, focusing on efficient sensor data management.
- Integrated an AI-driven optimization model to adjust runtime parameters, maximizing machine efficiency and carbon capture yield.

UNReLo (University of Nairobi) | Software Developer (2.5 years)

- Designed and implemented the end-to-end communication software linking a LoRa gateway to a satellite and then to a ground station.
- Mentored students on software configuration, usage, and best practices for establishing stable communication setups.

PROJECTS

SummarIze – Custom Summary Tool | React.js, Node.js, PayPal, RESTful APIs | https://summarize.cloud

- Built and deployed a browser extension that allows users to generate custom-formatted summaries from any website or local file.
- Designed and implemented a freemium business model with free usage tokens and a recurring subscription tier for premium features.
- Developed a secure backend server to manage user data, process API requests, and handle all payment transactions via the **PayPal API**.

Tic-tac-Toe Game AI Game | HTML, CSS, JavaScript | https://eugenemmf.github.io/erasure-tic-tac-toe

- Constructed a browser-based Tic-Tac-Toe game with a responsive UI and cookie-based session management for progress tracking.
- Implemented a configurable AI opponent using JavaScript, with varying difficulty levels based on game-tree algorithms.

AI Document Analysis Tool | React.js, Python, Gemini 1.5 Flash, VPS | https://github.com/EugeneMMF/sustain

- Led a team to architect a full-stack application for analyzing corporate sustainability reports from various file formats.
- Engineered a data processing pipeline using the Gemini 1.5 Flash API to extract insights and render them in graphical, tabular, and text formats.
- Managed collaborative development and version control using Git/GitHub and deployed the final solution on a Virtual Private Server (VPS).

Sudoku Solver from Image | Python, OpenCV, TeserractOCR, HTML, CSS | https://github.com/EugeneMMF/sudoku-solver

- Developed a full-stack solution that solves sudoku puzzles from an uploaded image in under 5 seconds.
- Leveraged image-based AI models for digit recognition and implemented advanced algorithms for solving the puzzle logic efficiently.

AI Development | Python | https://github.com/EugeneMMF/AI

- Developed a full AI library in Python, where the entire library was written from scratch using Numpy for faster ML computations.
- Optimized library performance by incorporating automatic CPU core-based acceleration.